

Foundations for Regime-Aware Cross-Market Equity Forecasting

Phase 1: HMM Regime Detection & Baseline Modeling

PROJECT TEAM

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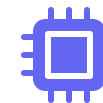
PROBLEM STATEMENT & MOTIVATION

The Core Challenge

- ⚠ Static market assumptions fail during regime shifts (crises vs. stable periods).
- 🌐 Cross-market forecasting (US & India) involves complex, non-stationary risk structures.

Our Objective

Build a robust foundation for regime-aware modeling before implementing advanced deep architectures.



PHASE 1 SCOPE: METHODOLOGY

STEP 01

Data Pipeline

- ✓ Acquisition of US (JPM) and Indian (NSE) stock data.
- ✓ Rigorous cleaning and preprocessing for daily OHLCV.

STEP 02

EDA & Diagnostics

- ✓ Distributional analysis for heavy-tail detection.
- ✓ Volatility clustering and regime identification.

STEP 03

Feature Engineering

- ✓ 36-column pipeline across 7 specialized groups.
- ✓ Technical, momentum, and cross-asset correlation.

STEP 04

Baseline Evaluation

- ✓ HMM-based regime detection (Bull/Bear/High-Vol).
- ✓ Walk-forward XGBoost evaluation framework.

DATA & CAUSALITY DESIGN

MULTI-MARKET ASSETS

Timeframe: 2010 – Present (Daily OHLCV)

 JPM (NYSE)	 ^GSPC (S&P 500)
 RELIANCE.NS	 ^NSEI (Nifty 50)
 TCS.NS	 HDFCBANK.NS
 EPIGRAL.NS	

5 tickers covering US Financials and Indian Energy, IT, Banking, and Small-Cap Chemicals.

CAUSALITY & INTEGRITY

T-1 Alignment: Applied to all cross-market features to ensure strict causality.

 **Zero Leakage:** Prevents look-ahead bias by synchronizing global market closing times.

 **Trustworthy Evaluation:** Foundation for reliable out-of-sample performance metrics.

EDA: HEAVY TAILS & RISK STRUCTURE

Non-Stationary Diagnostics

Our Exploratory Data Analysis reveals that equity returns across both US and Indian markets exhibit significant **heavy tails**.

With Kurtosis values far exceeding the Gaussian benchmark of 3, the risk of extreme price movements is statistically much higher than standard models assume.

KEY INSIGHT

Gaussian-world assumptions are empirically weak here; **regime-aware modeling** is fundamentally justified to capture these fat-tail risks.

Kurtosis Metrics

TICKER	KURTOSIS VALUE
JPM (US)	12.54
RELIANCE (IN)	9.13
TCS (IN)	6.77
HDFCBANK (IN)	10.28

EDA: VOLATILITY REGIMES

Volatility Clustering

20-day rolling volatility shows clear periods of relative calm punctuated by sudden, clustered spikes.

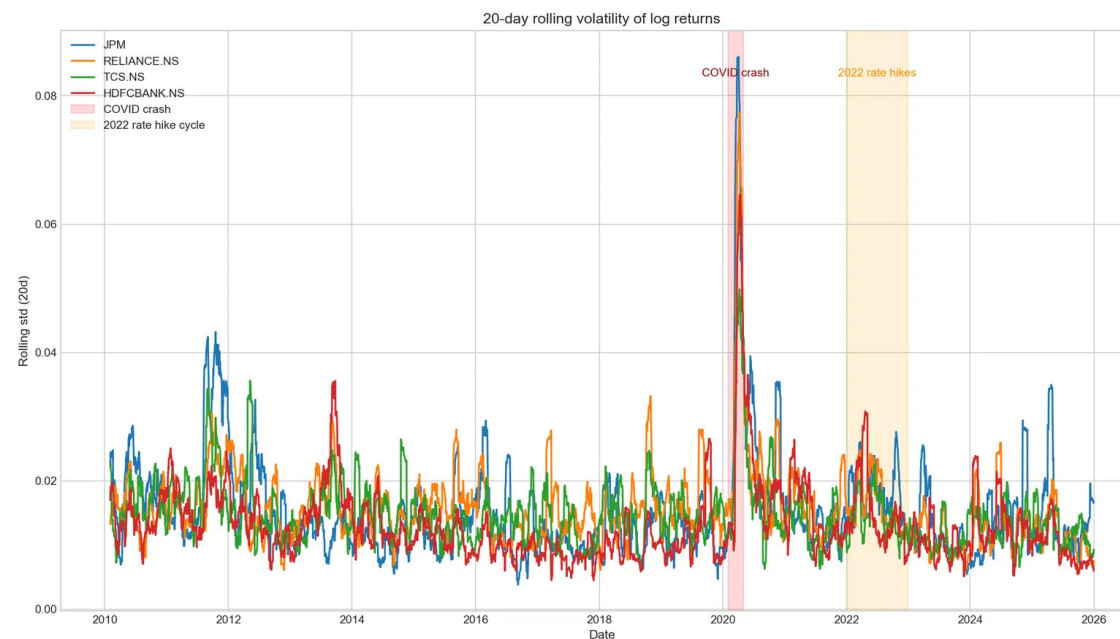
State-Dependency

Major market stress events visibly alter risk levels, confirming that a single static process is insufficient for modeling.

HMM Motivation

These distinct "risk states" justify the use of Hidden Markov Models to explicitly label market regimes.

Rolling Volatility (20-day)



Source: eda/plots/rolling_volatility.png

FEATURE ENGINEERING & HMM MODULE

36-COLUMN PIPELINE

-  **Lagged Returns:** Price memory (1-20 days)
-  **Volatility:** Garman-Klass, Vol-Ratio
-  **Technicals:** RSI, MACD, Bollinger
-  **Momentum:** ROC, Trend direction
-  **Volume:** Participation metrics
-  **HMM Regimes:** State-based labels
-  **Cross-Asset:** Rolling correlations

HMM REGIME DETECTION

Gaussian HMM trained on Log Returns and 20-day Realized Volatility.

$$a_{ij} = P(s_t = j \mid s_{t-1} = i)$$

$$b_j(x_t) = N(x_t; \mu_j, \Sigma_j)$$

BULL

BEAR

HIGH-VOL

Captures transition probabilities between market states to adapt feature weights dynamically.

BASLINE MODELING: XGBOOST

MODELING PROTOCOL

- 🎯 **Target:** Next-day log returns for all tickers.

Validation: Expanding-window walk-forward protocol to maintain chronological integrity.

- 🔲 **Optimization:** Early stopping implemented within each fold to prevent overfitting.

EVALUATION FRAMEWORK

- 📊 **Error Metrics:** RMSE and MAE for point-estimate precision.
- 📈 **Directional:** Directional Accuracy (DA%) for trend prediction.
- ⚖️ **Risk-Adjusted:** Sharpe Ratio to evaluate return vs. volatility.

Design Rationale: This framework mimics realistic deployment scenarios and avoids the "random-split optimism" common in financial time-series modeling.

PHASE 1 RESULTS: REGIME-STRATIFIED XGBOOST (FINAL CONFIRMED)

Ticker	RMSE	MAE	DA%	Sharpe
JPM	0.0194	0.0127	52.13	0.53
RELIANCE.NS	0.0180	0.0121	52.25	0.54
TCS.NS	0.0151	0.0106	53.17	0.85
HDFCBANK.NS	0.0165	0.0111	51.52	0.04
EPIGRAL.NS	0.0243	0.0166	51.32	-0.56
Average	0.0187	0.0126	52.08	0.28

How to read: Lower RMSE/MAE is better; higher DA% and Sharpe are better.

“Regime-stratified XGBoost achieves average DA of 52.08%. TCS shows strongest improvement with Sharpe 0.85. HDFCBANK and EPIGRAL show weak results — attributed to noisy HMM regime labels and small-cap idiosyncratic risk respectively.”

EPIGRAL-Specific Insight:

“EPIGRAL.NS was included as a small-cap generalization test. Negative Sharpe (-0.56) confirms that technical features alone are insufficient for small-cap stocks — motivating multimodal integration in RAMT Phase 2.”

Table 2 — Top 5 Features Per Ticker

JPM Momentum_60, BB_Lower, BB_Upper, Rolling_Corr_Index, Momentum_20	RELIANCE.NS Garman_Klass_Vol, MACD, Return_Lag_2, BB_Position, ROC_10	TCS.NS Momentum_20, BB_Upper, MACD_Hist, ROC_10, Momentum_60	HDFCBANK.NS Momentum_60, Momentum_20, HMM_Regime, BB_Lower, MACD	EPIGRAL.NS HMM_Regime, ROC_10, Momentum_20, MACD_Hist, Rolling_Corr_Index
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“HMM_Regime appears in top 5 for HDFCBANK and EPIGRAL, providing direct empirical evidence that regime labels carry predictive signal. Rolling_Corr_Index ranks in top 5 for JPM and EPIGRAL, validating the cross-asset correlation feature.”

WHAT PHASE 1 TAUGHT US → WHY PHASE 2



Baseline directional edge exists (average DA **52.08%**), but risk-adjusted gains are uneven.



Regime and cross-asset features show **predictive relevance** across multiple tickers.



Small-cap failure case (EPIGRAL) indicates **OHLCV-only limits** for volatile assets.



Phase 2 will integrate **multimodal signals** (news sentiment + Google Trends) with regime-aware RAMT.

CONCLUSION & PHASE 2 ROADMAP

PHASE 1 ACCOMPLISHMENTS

-  Validated leakage-safe data flow across US and Indian markets.
-  Implemented robust HMM-based regime detection and feature logic.
-  Established a credible performance baseline using walk-forward XGBoost.

PHASE 2 ROADMAP

-  Transition to RAMT-style regime-gated deep modeling (MoE Transformer).
- Deeper failure analysis categorized by specific market regimes.
-  Finalize unified explicit HMM-gated cross-market transfer architecture.

"We solved the foundation properly first, so later architecture claims will be credible."